



Data Mining: Concepts and Techniques

— Slides for Textbook —
— Chapter 1 —

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Acknowledgements

- This work on this set of slides started with my (Han's) tutorial for UCLA Extension course in February 1998
- Dr. [Hongjun Lu](#) from Hong Kong Univ. of Science and Technology taught jointly with me a Data Mining Summer Course in Shanghai, China in July 1998. He has contributed many excellent slides to it
- Some graduate students have contributed many new slides in the following years. Notable contributors include [Eugene Belchev](#), [Jian Pei](#), and [Osmar R. Zaiane](#) (now teaching in Univ. of Alberta).

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CMPT-459-00.3 Course Schedule

- Chapter 1. Introduction {W1:L2, L3}
- Chapter 2. Data warehousing and OLAP technology for data mining {W2:L1-3, W3:L1-2}
 - Homework # 1 distribution (SQLServer7.0+ DBMiner2.0)
- Chapter 3. Data preprocessing {W3:L3, W4: L1-L2}
- Chapter 4. Data mining primitives, languages and system architectures {W4: L3, W5: L1}
 - Homework #1 due, homework #2 distribution
- Chapter 5. Concept description: Characterization and comparison {W5: L2, L3, W6: L2}
 - W6:L1 Thanksgiving Day
- Chapter 6. Mining association rules in large databases {W6: L3, W7: L1-3, W8: L2}
 - Midterm {W8: L2}
- Chapter 7. Classification and prediction {W8:L3, W9: L1-L3}
- Chapter 8. Clustering analysis {W10: L1-L3}
 - W10: L3 Homework #2 due
- Chapter 9. Mining complex types of data {W11: L2-L3, W12:L1-L3}
 - W11:L1 Remembrance Day, W12:L3 Course project due
- Chapter 10. Data mining applications and trends in data mining {W13: L1-L3}
- Final Exam (W14)

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Where to Find the Set of Slides?

- Tutorial sections (MS PowerPoint files):
 - <http://www.cs.sfu.ca/~han/dmbook>
- Other conference presentation slides (.ppt):
 - <http://db.cs.sfu.ca/> or <http://www.cs.sfu.ca/~han>
- Research papers, DBMiner system, and other related information:
 - <http://db.cs.sfu.ca/> or <http://www.cs.sfu.ca/~han>

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Chapter 1. Introduction

- Motivation: Why data mining?
- What is data mining?
- Data Mining: On what kind of data?
- Data mining functionality
- Are all the patterns interesting?
- Classification of data mining systems
- Major issues in data mining

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Motivation: “Necessity is the Mother of Invention”

- Data explosion problem
 - Automated data collection tools and mature database technology lead to tremendous amounts of data stored in databases, data warehouses and other information repositories
- We are drowning in data, but starving for knowledge!
- Solution: Data warehousing and data mining
 - Data warehousing and on-line analytical processing
 - Extraction of interesting knowledge (rules, regularities, patterns, constraints) from data in large databases

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Evolution of Database Technology

(See Fig. 1.1)

- 1960s:
 - Data collection, database creation, IMS and network DBMS
- 1970s:
 - Relational data model, relational DBMS implementation
- 1980s:
 - RDBMS, advanced data models (extended-relational, OO, deductive, etc.) and application-oriented DBMS (spatial, scientific, engineering, etc.)
- 1990s—2000s:
 - Data mining and data warehousing, multimedia databases, and Web databases

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What Is Data Mining?



- Data mining (knowledge discovery in databases):
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) information or patterns from data in large databases
- Alternative names and their “inside stories”:
 - Data mining: a misnomer?
 - Knowledge discovery(mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- What is not data mining?
 - (Deductive) query processing.
 - Expert systems or small ML/statistical programs



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Why Data Mining? — Potential Applications

- Database analysis and decision support
 - Market analysis and management
 - target marketing, customer relation management, market basket analysis, cross selling, market segmentation
 - Risk analysis and management
 - Forecasting, customer retention, improved underwriting, quality control, competitive analysis
 - Fraud detection and management
- Other Applications
 - Text mining (news group, email, documents) and Web analysis.
 - Intelligent query answering

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Market Analysis and Management (1)

- Where are the data sources for analysis?
 - Credit card transactions, loyalty cards, discount coupons, customer complaint calls, plus (public) lifestyle studies
- Target marketing
 - Find clusters of “model” customers who share the same characteristics: interest, income level, spending habits, etc.
- Determine customer purchasing patterns over time
 - Conversion of single to a joint bank account: marriage, etc.
- Cross-market analysis
 - Associations/co-relations between product sales
 - Prediction based on the association information

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Market Analysis and Management (2)

- Customer profiling
 - data mining can tell you what types of customers buy what products (clustering or classification)
- Identifying customer requirements
 - identifying the best products for different customers
 - use prediction to find what factors will attract new customers
- Provides summary information
 - various multidimensional summary reports
 - statistical summary information (data central tendency and variation)

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Corporate Analysis and Risk Management

- Finance planning and asset evaluation
 - cash flow analysis and prediction
 - contingent claim analysis to evaluate assets
 - cross-sectional and time series analysis (financial-ratio, trend analysis, etc.)
- Resource planning:
 - summarize and compare the resources and spending
- Competition:
 - monitor competitors and market directions
 - group customers into classes and a class-based pricing procedure
 - set pricing strategy in a highly competitive market

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Fraud Detection and Management (1)

- Applications
 - widely used in health care, retail, credit card services, telecommunications (phone card fraud), etc.
- Approach
 - use historical data to build models of fraudulent behavior and use data mining to help identify similar instances
- Examples
 - auto insurance: detect a group of people who stage accidents to collect on insurance
 - money laundering: detect suspicious money transactions (US Treasury's Financial Crimes Enforcement Network)
 - medical insurance: detect professional patients and ring of doctors and ring of references

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Fraud Detection and Management (2)

- Detecting inappropriate medical treatment
 - Australian Health Insurance Commission identifies that in many cases blanket screening tests were requested (save Australian \$1m/yr).
- Detecting telephone fraud
 - Telephone call model: destination of the call, duration, time of day or week. Analyze patterns that deviate from an expected norm.
 - British Telecom identified discrete groups of callers with frequent intra-group calls, especially mobile phones, and broke a multimillion dollar fraud.
- Retail
 - Analysts estimate that 38% of retail shrink is due to dishonest employees.

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Other Applications

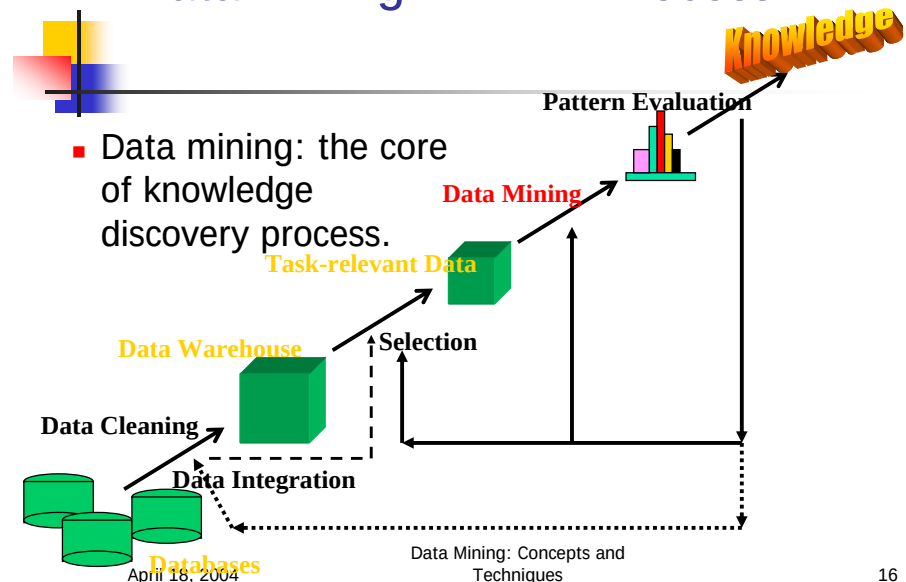
- Sports
 - IBM Advanced Scout analyzed NBA game statistics (shots blocked, assists, and fouls) to gain competitive advantage for New York Knicks and Miami Heat
- Astronomy
 - JPL and the Palomar Observatory discovered 22 quasars with the help of data mining
- Internet Web Surf-Aid
 - IBM Surf-Aid applies data mining algorithms to Web access logs for market-related pages to discover customer preference and behavior pages, analyzing effectiveness of Web marketing, improving Web site organization, etc.

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Data Mining: A KDD Process



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Steps of a KDD Process

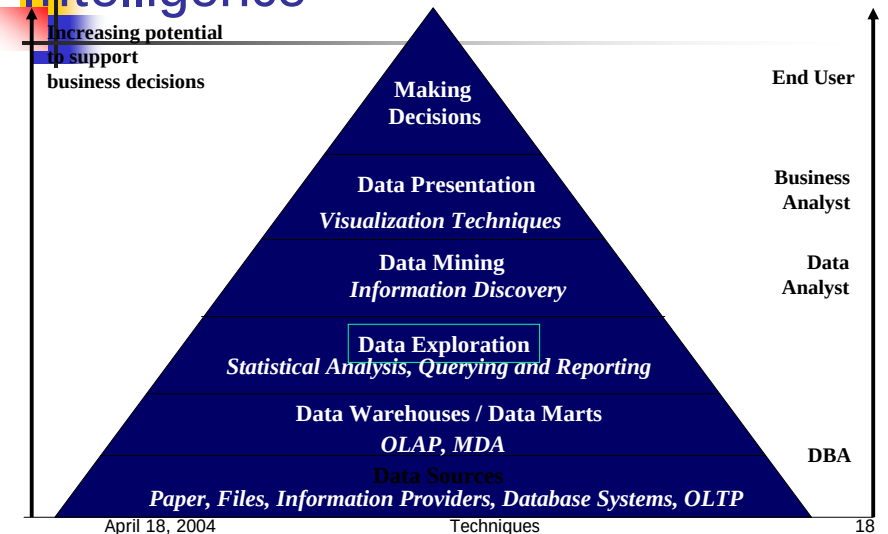
- Learning the application domain:
 - relevant prior knowledge and goals of application
- Creating a target data set: data selection
- **Data cleaning** and preprocessing: (may take 60% of effort!)
- **Data reduction and transformation**:
 - Find useful features, dimensionality/variable reduction, invariant representation.
- Choosing functions of data mining
 - summarization, classification, regression, association, clustering.
- Choosing the mining algorithm(s)
- **Data mining**: search for patterns of interest
- **Pattern evaluation and knowledge presentation**
 - visualization, transformation, removing redundant patterns, etc.
- Use of discovered knowledge

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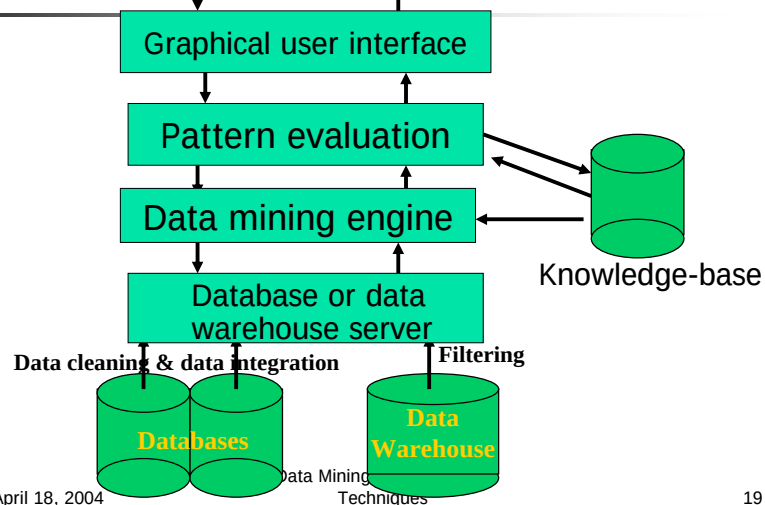
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Data Mining and Business Intelligence



Architecture of a Typical Data Mining System



Data Mining: On What Kind of Data?

- Relational databases
- Data warehouses
- Transactional databases
- Advanced DB and information repositories
 - Object-oriented and object-relational databases
 - Spatial databases
 - Time-series data and temporal data
 - Text databases and multimedia databases
 - Heterogeneous and legacy databases
 - WWW

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Data Mining Functionalities (1)

- Concept description: Characterization and discrimination
 - Generalize, summarize, and contrast data characteristics, e.g., dry vs. wet regions
- Association (correlation and causality)
 - Multi-dimensional vs. single-dimensional association
 - $\text{age}(X, "20..29") \wedge \text{income}(X, "20..29K") \rightarrow \text{buys}(X, "PC")$ [support = 2%, confidence = 60%]
 - $\text{contains}(T, "computer") \rightarrow \text{contains}(x, "software")$ [1%, 75%]

Data Mining Functionalities (2)

- Classification and Prediction
 - Finding models (functions) that describe and distinguish classes or concepts for future prediction
 - E.g., classify countries based on climate, or classify cars based on gas mileage
 - Presentation: decision-tree, classification rule, neural network
 - Prediction: Predict some unknown or missing numerical values
- Cluster analysis
 - Class label is unknown: Group data to form new classes, e.g., cluster houses to find distribution patterns
 - Clustering based on the principle: maximizing the intra-class similarity and minimizing the interclass similarity

Data Mining Functionalities (3)

- Outlier analysis
 - Outlier: a data object that does not comply with the general behavior of the data
 - It can be considered as noise or exception but is quite useful in fraud detection, rare events analysis
- Trend and evolution analysis
 - Trend and deviation: regression analysis
 - Sequential pattern mining, periodicity analysis
 - Similarity-based analysis
- Other pattern-directed or statistical analyses

Are All the “Discovered” Patterns Interesting?

- A data mining system/query may generate thousands of patterns, not all of them are interesting.
 - Suggested approach: Human-centered, query-based, focused mining
- **Interestingness measures:** A pattern is **interesting** if it is easily understood by humans, valid on new or test data with some degree of certainty, potentially useful, novel, or validates some hypothesis that a user seeks to confirm
- **Objective vs. subjective interestingness measures:**
 - Objective: based on statistics and structures of patterns, e.g., support, confidence, etc.
 - Subjective: based on user's belief in the data, e.g., unexpectedness, novelty, actionability, etc.

Can We Find All and Only Interesting Patterns?

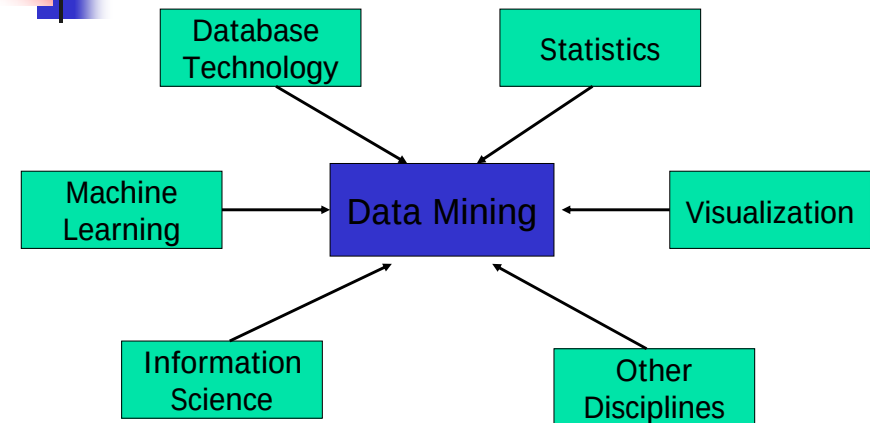
- Find all the interesting patterns: Completeness
 - Can a data mining system find all the interesting patterns?
 - Association vs. classification vs. clustering
- Search for only interesting patterns: Optimization
 - Can a data mining system find only the interesting patterns?
 - Approaches
 - First general all the patterns and then filter out the uninteresting ones.
 - Generate only the interesting patterns—mining query optimization

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Data Mining: Confluence of Multiple Disciplines



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Data Mining: Classification Schemes

- General functionality
 - Descriptive data mining
 - Predictive data mining
- Different views, different classifications
 - Kinds of databases to be mined
 - Kinds of knowledge to be discovered
 - Kinds of techniques utilized
 - Kinds of applications adapted

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A Multi-Dimensional View of Data Mining Classification

- **Databases to be mined**
 - Relational, transactional, object-oriented, object-relational, active, spatial, time-series, text, multi-media, heterogeneous, legacy, WWW, etc.
- **Knowledge to be mined**
 - Characterization, discrimination, association, classification, clustering, trend, deviation and outlier analysis, etc.
 - Multiple/integrated functions and mining at multiple levels
- **Techniques utilized**
 - Database-oriented, data warehouse (OLAP), machine learning, statistics, visualization, neural network, etc.
- **Applications adapted**
 - Retail, telecommunication, banking, fraud analysis, DNA mining, stock market analysis, Web mining, Weblog analysis, etc.

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OLAP Mining: An Integration of Data Mining and Data Warehousing

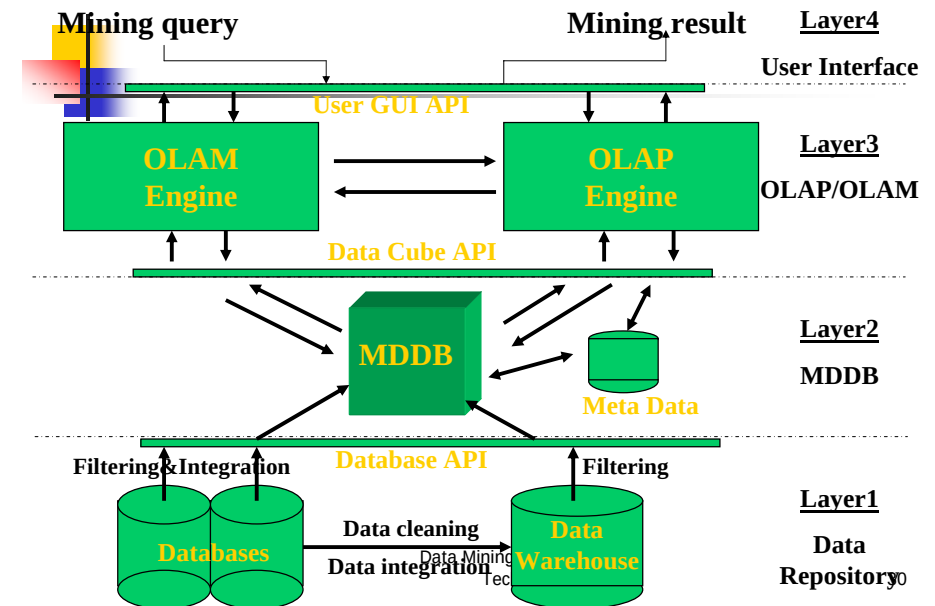
- **Data mining systems, DBMS, Data warehouse systems coupling**
 - No coupling, loose-coupling, semi-tight-coupling, tight-coupling
- **On-line analytical mining data**
 - integration of mining and OLAP technologies
- **Interactive mining multi-level knowledge**
 - Necessity of mining knowledge and patterns at different levels of abstraction by drilling/rolling, pivoting, slicing/dicing, etc.
- **Integration of multiple mining functions**
 - Characterized classification, first clustering and then association

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An OLAM Architecture



Major Issues in Data Mining (1)

- Mining methodology and user interaction
 - Mining different kinds of knowledge in databases
 - Interactive mining of knowledge at multiple levels of abstraction
 - Incorporation of background knowledge
 - Data mining query languages and ad-hoc data mining
 - Expression and visualization of data mining results
 - Handling noise and incomplete data
 - Pattern evaluation: the interestingness problem
- Performance and scalability
 - Efficiency and scalability of data mining algorithms
 - Parallel, distributed and incremental mining methods

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Major Issues in Data Mining (2)

- Issues relating to the diversity of data types
 - Handling relational and complex types of data
 - Mining information from heterogeneous databases and global information systems (WWW)
- Issues related to applications and social impacts
 - Application of discovered knowledge
 - Domain-specific data mining tools
 - Intelligent query answering
 - Process control and decision making
 - Integration of the discovered knowledge with existing knowledge: A knowledge fusion problem
 - Protection of data security, integrity, and privacy

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Summary

- Data mining: discovering interesting patterns from large amounts of data
- A natural evolution of database technology, in great demand, with wide applications
- A KDD process includes data cleaning, data integration, data selection, transformation, data mining, pattern evaluation, and knowledge presentation
- Mining can be performed in a variety of information repositories
- Data mining functionalities: characterization, discrimination, association, classification, clustering, outlier and trend analysis, etc.
- Classification of data mining systems
- Major issues in data mining

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A Brief History of Data Mining Society

- 1989 IJCAI Workshop on Knowledge Discovery in Databases (Piatetsky-Shapiro)
 - Knowledge Discovery in Databases (G. Piatetsky-Shapiro and W. Frawley, 1991)
- 1991-1994 Workshops on Knowledge Discovery in Databases
 - Advances in Knowledge Discovery and Data Mining (U. Fayyad, G. Piatetsky-Shapiro, P. Smyth, and R. Uthurusamy, 1996)
- 1995-1998 International Conferences on Knowledge Discovery in Databases and Data Mining (KDD'95-98)
 - Journal of Data Mining and Knowledge Discovery (1997)
- 1998 ACM SIGKDD, SIGKDD'1999-2001 conferences, and SIGKDD Explorations
- More conferences on data mining
 - PAKDD, PKDD, SIAM-Data Mining, (IEEE) ICDM, etc.

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Where to Find References?

- Data mining and KDD (SIGKDD member CDROM):
 - Conference proceedings: KDD, and others, such as PKDD, PAKDD, etc.
 - Journal: Data Mining and Knowledge Discovery
- Database field (SIGMOD member CD ROM):
 - Conference proceedings: ACM-SIGMOD, ACM-PODS, VLDB, ICDE, EDBT, DASFAA
 - Journals: ACM-TODS, J. ACM, IEEE-TKDE, JIIS, etc.
- AI and Machine Learning:
 - Conference proceedings: Machine learning, AAAI, IJCAI, etc.
 - Journals: Machine Learning, Artificial Intelligence, etc.
- Statistics:
 - Conference proceedings: Joint Stat. Meeting, etc.
 - Journals: Annals of statistics, etc.
- Visualization:
 - Conference proceedings: CHI, etc.
 - Journals: IEEE Trans. visualization and computer graphics, etc.

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References

- U. M. Fayyad, G. Piatetsky-Shapiro, P. Smyth, and R. Uthurusamy. Advances in Knowledge Discovery and Data Mining. AAAI/MIT Press, 1996.
- J. Han and M. Kamber. Data Mining: Concepts and Techniques. Morgan Kaufmann, 2000.
- T. Imielinski and H. Mannila. A database perspective on knowledge discovery. Communications of ACM, 39:58-64, 1996.
- G. Piatetsky-Shapiro, U. Fayyad, and P. Smith. From data mining to knowledge discovery: An overview. In U.M. Fayyad, et al. (eds.), Advances in Knowledge Discovery and Data Mining, 1-35. AAAI/MIT Press, 1996.
- G. Piatetsky-Shapiro and W. J. Frawley. Knowledge Discovery in Databases. AAAI/MIT Press, 1991.

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<http://www.cs.sfu.ca/~han>



Thank you !!!

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CMPT-843 Course Arrangement

- 1st week: full instructor teaching
- 2nd to 11th week: 1/2 graduate student + 1/2 instructor teaching
- 12-13th week: full student graduate project presentation
- Course evaluation:
 - presentation (quality of presentation slides 7% + presentation 8%) 15%
 - midterm exam 35%
 - project (presentation 5% + report 25%) total 30%
 - homework (2): 20%
- Deadline for the selection of your work in the semester:
 - selection of course presentation: at the end of the 1st week
 - selection of the course project: at the end of the 3rd week
 - project proposal due date: at the end of the 4th week
 - homework due dates: Data Mining: Concepts and Techniques
 - project due date: end of the semester

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